



Cubic Spline Model To Predict COVID-19 Mortality Based On Age

for the Bachelor of Science Degree in Mathematics

By

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Declaration

I, D.K.S.Kothalawala, hereby declare that the project report titled, "Cubic Spline Model To Predict COVID-19 Mortality Based On Age" submitted to Department of Mathematics, Faculty of Science, University of Ruhuna, in partial fulfillment of the requirements for the Level I course unit Mathematical Computing of the degree of Bachelor of Science in Faculty of Science, University of Ruhuna. It has not been submitted to any other institution or study program.

I declare that this report is the result of my own independent work under the supervision of Dr.E.J.K.P.Nandani.

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Date : 2024.09.20

Supervisor's Certificate

This is to certify that the project report entitled "Cubic Spline Model To Predict COVID-19 Mortality Based On Age" is a record of the work under my guidance and supervision.

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Abstract

The project presents about the theory of interpolation. It specially focused on cubic spline interpolation. Cubic spline interpolation is a mathematical method that is utilized to interpolate a continuous curve over some specified data points. Simply put, a series of cubic polynomials (or splines) is established between specified data points. The conditions enforced ensure that the resulting spline becomes a continuous polynomial function across each of the data points. An essential component of cubic splines' definition is that they possess continuous first and second derivatives everywhere within the spline function. A numerical example of cubic spline interpolation in the case of boundary conditions is also given. This project discussed calculating the number of COVID-19 deaths by age using cubic spline interpolation.

Keywords : Data interpolation, Linear spline, Quadratic spline, Cubic spline, Polynomial spline, Continuity, Degree.

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Chapter 1

Introduction

The coronavirus, more specifically SARS-CoV-2 is a new type of virus that first came to the public's attention late in the year 2019 and became the cause of the COVID-19 illness. It is transmitted mostly through droplets formed during respiratory and has complicated symptoms that can manifest in moderate form and extreme form of the disease. In severe cases, especially if the client is elderly, or has other health issues they may experience life-threatening consequences of the disease. In view of this, there were various measures put in place due to the pandemic such as policies concerning wearing of masks, lock down and vaccination among others.

The current COVID-19, which is caused by the COVID-19 virus, has been a global burden, with millions of cases and deaths recorded worldwide. An important predictor of mortality from COVID-19 is the people's age. Cross-country analysis of COVID-19 data has revealed that as people grow older, they are at higher risks of serious illness and death than the younger generation. This trend is also attributable to the biological and health-related risks associated with aging. As people grow older, they have poor or compromised immune systems that make it challenging to fight infections. Further, the elderly are vulnerable to developing serious effects from COVID-19 because they are more prone to the development of chronic diseases such as heart complications, diabetes, and chronic respiratory illnesses. These conditions, known as the co-morbid conditions, together with the other factors are very significant as far as the elderly are concerned. On the other hand, you have younger population sub-groups, particularly children and adolescents, who tend to have less severe symptoms and lower mortality from the disease. That being said, they are not completely protected from getting the virus, as serious, albeit quite rare, conditions such as MIS-C were reported. It is crucial to know the relation of age factor with COVID-19 mortality in

order to develop future plans regarding vaccine distribution, treatment access for high-risk groups, and precautionary strategies to be taken for high-risk groups. The findings stress the need for safeguarding elderly people with relevant measures, including taking vaccines during the early years and concentrating health services for them. This age-based mortality pattern has far-reaching implications for health care delivery systems, policies, and international strategies to combat the pandemic, with an emphasis on vulnerable groups.

1.1 Background of the study

Spline interpolation is a mathematical method utilized to create a parametric approximation of a smoothly flowing curve that passes through a set of points. Spline interpolation is helpful when we are trying to represent data using a single polynomial because one of the drawbacks of polynomial interpolation is that polynomials can oscillate between data points and provide approximations that lack validity. The term "spline interpolation" refers to a form of interpolation in which the piecewise polynomial is a specific type known as a spline interpolation. Interpolation can also be described as the process to determine the unknown values that lie in between the known data. Interpolation is often applied to predict unknown values for different related data points, which often relate to different geospatial data points such as noise level, rainfall, elevation, etc.

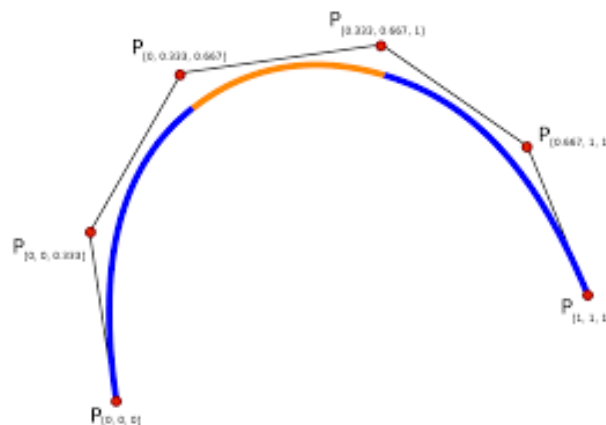


Figure 1.1: - Spline interpolation

1.1.1 Types of Spline Interpolation

Different types of spline interpolation polynomials vary based on their degrees, continuity and boundary conditions. Some of them are mentioned below.

- Linear spline, Quadratic spline, Cubic spline, Basis spline, Natural spline, Clamped spline etc.

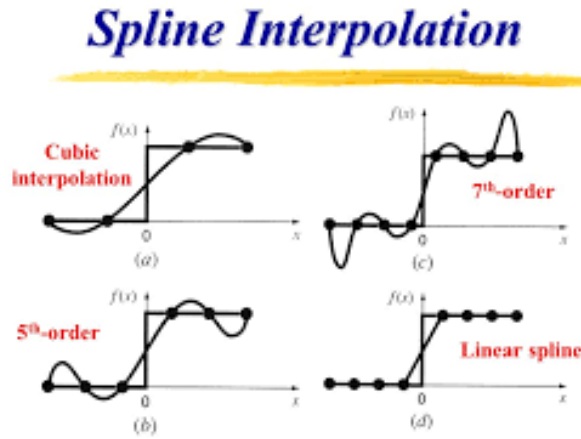


Figure 1.2: - Different types of spline interpolation

Linear spline, Quadratic spline and Cubic spline are defined below.

1.1.1.1 Linear Spline Interpolation

Linear spline interpolation relies on the data for neighboring data points and does not account for any data from other data points. Because it is distinct and is, by definition, close to neighboring data points, the slope of the spline does not change abruptly, indicating the first derivative at those data points is stably continuous. For $n = 1$ to n , you have access to an entire collection of data points as well as known data points (x_i, y_i) . Linear spline interpolation associates each nearby point pair with a straight line to get in between. The linear interpolation thus essentially happens on connected straight lines and contains linear equations for pairs (x_i, y_i) and (x_{i+1}, y_{i+1}) . Consequently, for each $[x_i, x]$, we may characterize the linear interpolation function $S_i(x)$.

$$S_i(x) = y_i + \frac{y_{i+1} - y_i}{x_{i+1} - x_i}(x - x_i) \quad (1.1)$$

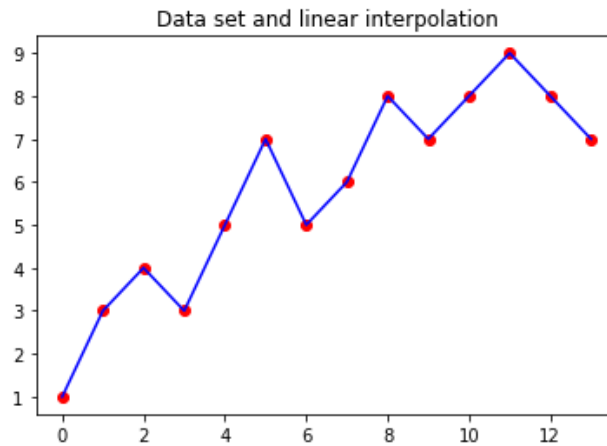


Figure 1.3: - Linear spline interpolation

1.1.1.2 Quadratic Spline Interpolation

Quadratic spline interpolation represents one strategy for tying data of different values together by bending a piecewise quadratic approximation curve of the data between two values. Interpolation intends to create an estimate $f(x)$, or a function of x , calculated from associated and known point data. Quadratic interpolation is a way of estimating the value of the slope of a curve to improve the estimate of our data; in both cases (quadratic spline and quadratic interpolation), we provide curvature to the line joining the points. The following conditions must exist in the quadratic spline:

1. At supporting points x_j and x_{i+1} , the spline matches the functional value.
2. The first derivative on the spline, at $x_i + 1$, is equivalent to the first derivative of S_i and S_{i+1} .
3. The first derivative should vanish at the lower boundary implementation.

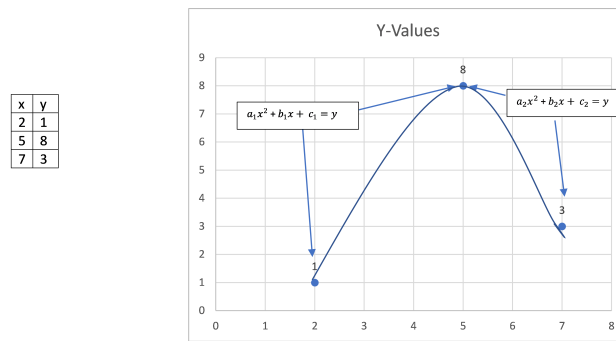


Figure 1.4: - Quadratic spline interpolation

1.1.1.3 Cubic spline interpolation

Cubic spline interpolation represents an exceptional instance of spline interpolation. Essentially, cubic splines enable us to select a cubic or lower degree curve to fit data points. The resultant interpolating polynomial has been established through the literature to be smoother and assure less error than lagrange polynomials and Newton polynomials. In this way, cubic splines are preferential over alternative interpolating polynomials based on smoothness and reduced error.

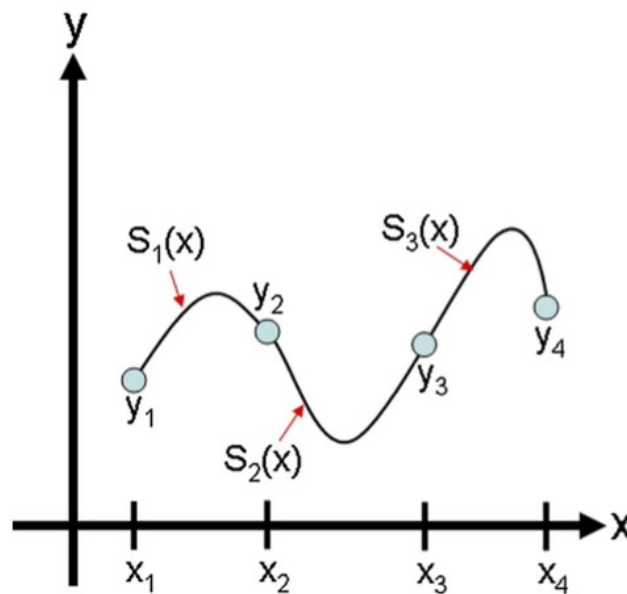


Figure 1.5: - Cubic spline interpolation

1.2 Problem Statement

COVID-19 is known to impact people of different ages. Higher mortality rates are expected to be recorded among groups of people belonging to older age classes. But, to comprehend this, it is not very difficult. They also have no ability to capture the patterns of the system as compared to the traditional models. Therefore, it proposed a cubic spline model. This model predicts mortality, with age being a factor. It produces a very aesthetic curve. It also incorporates flexibility and changes with respect to age. As our approach suggests, this increases the accuracy of mortality predictions. It facilitates making good plans that, if used, will result in productive outcomes.

1.3 Objectives

The objectives of this research study are to,

- Utilizing a cubic spline regression model to characterize the mortality rate in persons diagnosed with COVID-19 while age is a predictor.
- Utilizing cubic splines to analyze and model the nonlinear functional relationships of age to COVID-19 mortality and taking into consideration that the rates at which mortality occurs can be astonishingly different across age groups.
- Clearly identifying, if not clinically relevant, age cutoffs or cut-off points at which the risk of death is significantly increased to help define a high-risk age group.
- Investigate how death rates continue to increase across different age ranges instead of assuming a constant change in large and broad age groups.

1.4 Structure of the report

There are six sections in this report. Some background information and context for the research are given in the introduction. The literature review is an overview of previous research on the subject of our topic inquiry. Methodology examines the procedures or strategies used to gather and examine data. The study's findings are presented in the Results section. The results of the study are analyzed in the discussion section, as well as

the similarities and differences with the literature highlighted in the literature review. The conclusion section summarizes the findings and offers some recommendations based on the analysis. The C code related to the research is included at the end of the report.

Chapter 2

Literature Review

Estimating excess mortality due to the COVID-19 pandemic: a systematic analysis of COVID-19-related mortality, has been conducted in 2020–21

Mortality rates are a fundamental principle for public health decisions. Mortality varies temporally and spatially, and the measurement of mortality is subject to well-known biases that have been heightened by the experience of the COVID-19 pandemic. This report attempts to estimate excess mortality due to the COVID-19 pandemic across the dimensions of 191 countries and territories and 252 sub-national geographical units for selected countries from June 1, 2020, to December 31, 2021.

Prasad Chathuranga Weerasinghe ,Indumini Gunatilake , Nirma Alpitiarachchi , Indika Gunawardana , Ahmed Shiyam , Suwani Dasanayake , Gayani Dissanayake , Thanuja Ratnayaka , Irshad Mashood , Disala Welgama , Jinadari Amarasena , Iresha Jayawickrama , Charith Hettiarachchi , RuwanFerdinando , Mahendra Arnold (2020) Conducted research on Measures of mortality in COVID-19: beyond the death toll.

Many epidemiological mortality rates, such as asymptomatic infection rate, case fatality rate, mortality rate, infection fatality rate, and excess mortality rate, have been widely used to capture the global impact of the COVID-19 pandemic in science and non-science publications. Each measure has limitations. Some indicators are sensitive to time and case fatality rates—for example, the CFR, which is highly sensitive to case data and may result in extreme values, particularly where the number of patients is low. Due to the possibility that there may have been instances of infection that were not diagnosed, infection mortality ratio is a good substitute for mortality rate. It is important to avoid misinterpretation of mortality measures by relaying and explaining those measures with some information,

which includes the following: These various epidemiological mortality measures have been widely employed in scientific articles as well as literature sources to generalize the global impact of the COVID-19 pandemic. Each measure has limitations. Some indicators are sensitive to time or case fatality rates, including the case fatality ratio, which may take very high or very low values and specifically when the number of cases is low. The infection mortality ratio can be considered as a suitable replacement for the mortality rate as, perhaps, there were some cases that were not diagnosed that could be useful for the estimate. [4]

Carukshi Arambepola, Nuwan Darshana Wickramasinghe, Surangi Jayakody, Sumudu Avanthi Hewage, Ananda Wijewickrema, Nalika Gunawardena, Sapumal Dhanapala, Shamini Prathapan (2021) conducted a research on Sri Lanka's early success in the containment of COVID-19 through its rapid response.

A cross-sectional study was performed using secondary data, collecting data from hospital records and investigation reports collected with all laboratory-investigated patients admitted to the three main state-sector hospitals in Sri Lanka from 11 March to 7 April, 2020. The clinical outcomes were classified into levels I-III on the basis of international standard grading of clinical severity. Kruskal-Wallis H, Mann Whitney U, and Fisher's exact tests were used for analysis of differences between the variables.[1]

Rubina Mulchandani, Giridhara R Babu, Avinash Kaur, Ranjana Singh, Tanica Lyngdoh (2022) conducted a research Factors associated with differential COVID-19 mortality rates in the SEAR nations.

In December 2019, the world was introduced to a very dangerous virus, which is known as coronavirus (COVID-19), that has resulted in massive loss of lives, dilation of the health sector, and distortion in the global economy. High-income countries have also received increased attention in terms of chronic mortality trends. Nevertheless, comparable data from L/MINs are comparatively scarce. In this analysis, they sought to present the trend of COVID-19 mortality in WHO-SEAR nations and identify factors that might be used to justify such trends.[2]

example one[3]

Chapter 3

Methodology

3.1 Cubic spline interpolation

The purpose of cubic spline interpolation is just to get an equation that can be used for finite interpolation. continuous at the first and second derivatives at all the intervals and also continuous at the initial points. interpolating nodes. This will help us to have a better interpolating function, which in turn will help us to get a better approximation. Since the first derivative, $S'(x)$, is continuous, this means that the graph of $y = s(x)$ cannot make sudden jumps or corners and therefore no sharp corners. The fact that the second derivative does not jump states that the radius of curvature is given at each part of the curve.

3.2 Method

The equation of the cubic spline in the i^{th} interval $[x_{i-1}, x]$, is show in the following equation

$$s_i(x) = a_i x^3 + b_i x^2 + c_i x + d_i \quad (3.1)$$

where, a_i, b_i, c_i, d_i represents the 4n coefficient for $i=1,2, \dots n$

It is particularly noteworthy that the cubic spline interpolation in (3.1) is in conformity with the following conditions.

$$s_i(x_i) = s_i \quad (3.2)$$

$$s_i(x_{i+1}) = s_{i+1}(x_{i+1}) \quad (3.3)$$

$$S'_i(x_{i+1}) = S'_{i+1}(x_{i+1}) \quad (3.4)$$

$$S''_i(x_{i+1}) = S''_{i+1}(x_{i+1}) \quad (3.5)$$

Equation (3.2), together with (3.3), (3.4), and (3.5), provides a total of $4-2n$ conditions. When it comes to creating cubic spline interpolation, we require: Here, it is oral that at least $4n$ parameters need to be determined. In this paper will be used the natural boundary conditions. The net earnings formula is given below:

$$S''_0(x_0) = S''_{n-1}(x_n) = 0 \quad (3.6)$$

Since there are seven data points, there are six intervals, so we get six $s_i(x)$ equations.

$$s_1(x) = a_1x^3 + b_1x^2 + c_1x + d_1, [35, 45] \quad (3.7)$$

$$s_2(x) = a_2x^3 + b_2x^2 + c_2x + d_2, [45, 55] \quad (3.8)$$

$$s_3(x) = a_3x^3 + b_3x^2 + c_3x + d_3, [55, 65] \quad (3.9)$$

$$s_4(x) = a_4x^3 + b_4x^2 + c_4x + d_4, [65, 75] \quad (3.10)$$

$$s_5(x) = a_5x^3 + b_5x^2 + c_5x + d_5, [75, 85] \quad (3.11)$$

$$s_6(x) = a_6x^3 + b_6x^2 + c_6x + d_6, [85, 95] \quad (3.12)$$

In relation to conditions (3.2) and (3.3) .

$$s_1(35) = 2.4 \quad (3.13)$$

$$s_1(45) = 12.4 \quad (3.14)$$

$$s_2(45) = 12.4 \quad (3.15)$$

$$s_2(55) = 16.7 \quad (3.16)$$

$$s_3(55) = 16.7 \quad (3.17)$$

$$s_3(65) = 21.4 \quad (3.18)$$

$$s_4(65) = 21.4 \quad (3.19)$$

$$s_4(75) = 26.7 \quad (3.20)$$

$$s_5(75) = 26.7 \quad (3.21)$$

$$s_5(85) = 15.7 \quad (3.22)$$

$$s_6(85) = 15.7 \quad (3.23)$$

$$s_6(95) = 2.4 \quad (3.24)$$

$$2.4 = a_1 35^3 + b_1 35^2 + c_1 35 + d_1 \quad (3.25)$$

$$12.4 = a_1 45^3 + b_1 45^2 + c_1 45 + d_1 \quad (3.26)$$

$$12.4 = a_2 45^3 + b_2 45^2 + c_2 45 + d_2 \quad (3.27)$$

$$16.7 = a_2 55^3 + b_2 55^2 + c_2 55 + d_2 \quad (3.28)$$

$$16.7 = a_3 55^3 + b_3 55^2 + c_3 55 + d_3 \quad (3.29)$$

$$21.4 = a_3 65^3 + b_3 65^2 + c_3 65 + d_3 \quad (3.30)$$

$$21.4 = a_4 65^3 + b_4 65^2 + c_4 65 + d_4 \quad (3.31)$$

$$26.7 = a_4 75^3 + b_4 75^2 + c_4 75 + d_4 \quad (3.32)$$

$$26.7 = a_5 75^3 + b_5 75^2 + c_5 75 + d_5 \quad (3.33)$$

$$15.7 = a_5 85^3 + b_5 85^2 + c_5 85 + d_5 \quad (3.34)$$

$$15.7 = a_6 85^3 + b_6 85^2 + c_6 85 + d_6 \quad (3.35)$$

$$2.4 = a_6 95^3 + b_6 95^2 + c_6 95 + d_6 \quad (3.36)$$

In relation to conditions (3.4) and (3.5) .

By the continuity of the first derivative

$$S'_i(x) = 3a_i x^2 + 2b_i x + c_i \quad (3.37)$$

$$S'_1(45) = S'_2(45) \quad (3.38)$$

$$S'_2(55) = S'_3(55) \quad (3.39)$$

$$S'_3(65) = S'_4(65) \quad (3.40)$$

$$S'_4(75) = S'_5(75) \quad (3.41)$$

$$S'_5(85) = S'_6(85) \quad (3.42)$$

$$3a_145^2 + 2b_145 + c_1 = 3a_245^2 + 2b_245 + c_2 \quad (3.43)$$

$$3a_255^2 + 2b_255 + c_2 = 3a_355^2 + 2b_355 + c_3 \quad (3.44)$$

$$3a_365^2 + 2b_365 + c_3 = 3a_465^2 + 2b_465 + c_4 \quad (3.45)$$

$$3a_475^2 + 2b_475 + c_4 = 3a_575^2 + 2b_575 + c_5 \quad (3.46)$$

$$3a_585^2 + 2b_585 + c_5 = 3a_685^2 + 2b_685 + c_6 \quad (3.47)$$

By the continuity of the second derivative

$$S''_i(x) = 6a_ix + 2b_i \quad (3.48)$$

$$S''_1(45) = S''_2(45) \quad (3.49)$$

$$S''_2(55) = S''_3(55) \quad (3.50)$$

$$S''_3(65) = S''_4(65) \quad (3.51)$$

$$S''_4(75) = S''_5(75) \quad (3.52)$$

$$S''_5(85) = S''_6(85) \quad (3.53)$$

$$6a_145 + 2b_1 = 6a_245 + 2b_2 \quad (3.54)$$

$$6a_255 + 2b_2 = 6a_355 + 2b_3 \quad (3.55)$$

$$6a_365 + 2b_3 = 6a_465 + 2b_4 \quad (3.56)$$

$$6a_475 + 2b_4 = 6a_575 + 2b_5 \quad (3.57)$$

$$6a_585 + 2b_5 = 6a_685 + 2b_6 \quad (3.58)$$

Since the second derivative is zero at the last points

$$S''_1(35) = 0 \quad (3.59)$$

$$S''_6(95) = 0 \quad (3.60)$$

$$6a_135 + 2b_1 = 0 \quad (3.61)$$

$$6a_695 + 2b_6 = 0 \quad (3.62)$$

Insert the values of a,b, c and d of the twenty-four equations into a matrix and find the

values of a, b, c and d of the six equations by solving matrix.

The values obtained after solving the matrix are as follows

$$a_1 = -0.0015, b_1 = 0.1537, c_1 = -4.2332, d_1 = 25.0389$$

$$a_2 = 0.0016, b_2 = -0.2625, c_2 = 14.4964, d_2 = -255.9064$$

$$a_3 = 0.0011, b_3 = -0.1747, c_3 = 9.6658, d_3 = -167.3438$$

$$a_4 = -0.0058, b_4 = 1.1618, c_4 = -77.208, d_4 = 1714.8938$$

$$a_5 = 0.0051, b_5 = -1.2789, c_5 = 105.8481, d_5 = -2861.4764$$

$$a_6 = -0.0006, b_6 = 0.1585, c_6 = -16.3323, d_6 = 600.2989$$

The resulting equations are as follows

$$s_1(x) = -0.0015x^3 + 0.1537x^2 - 4.2332x + 25.0389, [35, 45] \quad (3.63)$$

$$s_2(x) = 0.0011x^3 - 0.2625x^2 + 14.4964x - 255.9064, [45, 55] \quad (3.64)$$

$$s_3(x) = 0.0011x^3 - 0.1747x^2 + 9.6658x - 167.3438, [55, 65] \quad (3.65)$$

$$s_4(x) = -0.0058x^3 + 1.1618x^2 - 77.2068x + 1714.8938, [65, 75] \quad (3.66)$$

$$s_5(x) = 0.0051x^3 - 1.2789x^2 + 105.8481x - 2861.4764, [75, 85] \quad (3.67)$$

$$s_6(x) = -0.0006x^3 + 0.1585x^2 - 16.3323x + 600.2989, [35, 45] \quad (3.68)$$

Let (x_i, y_i) , $i = 0, 1, \dots, n$ be a set of $(n+1)$ points, where $y = f(x)$. A cubic spline is a set of n third degree polynomials and each polynomial is constructed in each interval $[x_i, x_{i+1}]$, $i = 0, 1, \dots, n-1$.

The function $S(x)$ is called cubic spline in $[x_0, x_n]$.

Let the cubic spline on the $(i + 1)^{th}$ interval be

$$S_i(x) = A_i(x - x_i)^3 + B_i(x - x_i)^2 + C_i(x - x_i) + D_i \quad (3.69)$$

where A_i, B_i, C_i and D_i are unknown and their values are to be determined

Since it passes through the end points x_i and x_{i+1} , therefore,

$$S_i(x_i) = y_i = D_i \quad (3.70)$$

$$S_i(x_{i+1}) = y_{i+1} \quad (3.71)$$

and

$$y_{i+1} = A_i(x_{i+1} - x_i)^3 + B_i(x_{i+1} - x_i)^2 + C_i(x_{i+1} - x_i) + D_i \quad (3.72)$$

denoted by $h_{i+1} = x_{i+1} - x_i$, $i = 0, 1, 2, \dots, n-1$, therefore,

$$y_{i+1} = A_i h_{i+1}^3 + B_i h_{i+1}^2 + C_i h_{i+1} + D_i \quad (3.73)$$

Now, differentiate the equation (3.7) twice and get the following equations.

$$S'_i(x) = 3A_i(x - x_i)^2 + 2B_i(x - x_i) + C_i \quad (3.74)$$

and

$$S''_i(x) = 6A_i(x - x_i) + 2B_i \quad (3.75)$$

From the relationship represented by equation (3.13),

$$y''_i = S''(x_i) = 2B_i \text{ and } y''_{i+1} = S''(x_{i+1}) = 6A_i h_{i+1} + 2B_i$$

For convenience, we write y''_i by M_i for $i = 0, 1, 2, \dots, n$.

In this notation, the relationship rewrites as $M_i = 2B_i$, $M_i = 6A_i h_{i+1} + 2B_i$ Thus,

$$B_i = \frac{M_i}{2} \quad (3.76)$$

$$A_i = \frac{M_{i+1} - M_i}{6h_{i+1}} \quad (3.77)$$

Thus the constants A_i and B_i are represented in terms of M_i . Replacing A_i , B_i into equation (3.11), we obtain

$$y_{i+1} = \frac{M_{i+1} - M_i}{6h_{i+1}} h_{i+1}^3 + \frac{M_i}{2} h_{i+1}^2 + C_i h_{i+1} + y_i \quad (3.78)$$

That is

$$C_i = \frac{y_{i+1} - y_i}{h_{i+1}} - \frac{2h_{i+1}M_i + h_{i+1}M_i}{6} \quad (3.79)$$

Thus, we obtain c_i .

We now have all coefficients A_i , B_i , C_i , and D_i for $i = 0, 1, \dots, n-1$ from (5.5) in terms of $n+1$ unknowns $M_0, M_1, \dots, M_n$.

The next step is to these unknowns, and to do this we take the from (3.4); i.e., $S'_{i-1}(x_i) =$

$sS'_i(x_i)$ for $i = 1, 2, \dots, n-1$.

For the polynomial $S_i(x)$

$$S'_i(x_i) = 3A_i(x_i - x_i)^2 + 2B_i(x_i - x_i) + C_i = C_i, \quad (3.80)$$

and for the polynomial $S_{i-1}(x)$

$$S'_{i-1}(x_i) = 3A_{i-1}(x_i - x_{i-1})^2 + 2B_{i-1}(x_i - x_{i-1}) + C_{i-1}, \quad (3.81)$$

Now from the equation (3.18) and (3.19), we get

$$c_i = 3A_{i-1}h_i^2 + 2B_{i-1}h_i + C_{i-1}. \quad (3.82)$$

We substitute the values A_{i-1} , B_{i-1} , C_{i-1} , and C_i to the above equation and we get

$$\frac{y_{i+1} - y_i}{h_{i+1}} - \frac{2h_{i+1}M_i + h_{i+1}M_{i+1}}{6} = 3\left(\frac{M_i - M_{i-1}}{6h_i}\right)h_i^2 + \left(\frac{M_i}{2}\right)h_i + \frac{y_i - y_{i-1}}{h_i} - \frac{2h_iM_{i-1} + h_iM_i}{6}. \quad (3.83)$$

This gives

$$h_iM_{i-1} + 2(h_i + h_{i+1})M_i + h_{i+1}M_{i+1} = 6\left(\frac{y_{i+1} - y_i}{h_{i+1}} - \frac{y_i - y_{i-1}}{h_i}\right). \quad (3.84)$$

Where $i = 1, 2, \dots, n-1$. Let $a_i = h_i$, $b_i = 2(h_i + h_{i+1})$, $c_i = h_{i+1}$ and

$$d_i = 6\left(\frac{y_{i+1} - y_i}{h_{i+1}} - \frac{y_i - y_{i-1}}{h_i}\right).$$

Using these notations, the equation (3.22) reduces to

$$a_iM_{i-1} + b_iM_i + c_iM_{i+1} = d_i, \quad i = 1, 2, \dots, n-1 \quad (3.85)$$

The cubic spline formula is given by[?]:

$$f(x) = \frac{(x_i - x)^3}{6h}M_{i-1} + \frac{(x - x_{i-1})^3}{6h}M_i + \frac{(x_i - x)}{h}\left(y_{i-1} - \frac{h^2}{6}M_{i-1}\right) + \frac{(x - x_{i-1})}{h}\left(y_i - \frac{h^2}{6}M_i\right) \quad (3.86)$$

where $h = 10$ and $n = 6$.

The conditions are given by:

$$M_{i-1} + 4M_i + M_{i+1} = \frac{6}{h^2} (y_{i-1} - 2y_i + y_{i+1}) \quad (3.87)$$

With boundary conditions $M_0 = 0$ and $M_6 = 0$, we substitute $i = 1$ into equation (3.87):

$$M_0 + 4M_1 + M_2 = \frac{6}{h^2} (y_0 - 2y_1 + y_2) \quad (3.88)$$

$$0 + 4M_1 + M_2 = \frac{6}{100} (2.4 - 2 \cdot 12.4 + 16.7) \quad (3.89)$$

$$4M_1 + M_2 = -0.342 \quad (3.90)$$

Substitute $i = 2$ into equation (3.87):

$$M_1 + 4M_2 + M_3 = \frac{6}{h^2} (y_1 - 2y_2 + y_3) \quad (3.91)$$

$$M_1 + 4M_2 + M_3 = \frac{6}{100} (12.4 - 2 \cdot 16.7 + 21.4) \quad (3.92)$$

$$M_1 + 4M_2 + M_3 = 0.024 \quad (3.93)$$

Substitute $i = 3$ into equation (3.87):

$$M_2 + 4M_3 + M_4 = \frac{6}{h^2} (y_2 - 2y_3 + y_4) \quad (3.94)$$

$$M_2 + 4M_3 + M_4 = \frac{6}{100} (16.7 - 2 \cdot 21.4 + 26.7) \quad (3.95)$$

$$M_2 + 4M_3 + M_4 = 0.036 \quad (3.96)$$

Substitute $i = 4$ into equation (3.87):

$$M_3 + 4M_4 + M_5 = \frac{6}{h^2} (y_3 - 2y_4 + y_5) \quad (3.97)$$

$$M_3 + 4M_4 + M_5 = \frac{6}{100} (21.4 - 2 \cdot 26.7 + 15.7) \quad (3.98)$$

$$M_3 + 4M_4 + M_5 = -0.978 \quad (3.99)$$

Substitute $i = 5$ into equation (3.87):

$$M_4 + 4M_5 + M_6 = \frac{6}{h^2} (y_4 - 2y_5 + y_6) \quad (3.100)$$

$$M_4 + 4M_5 + 0 = \frac{6}{100} (26.7 - 2 \cdot 15.7 + 2.4) \quad (3.101)$$

$$M_4 + 4M_5 = -0.138 \quad (3.102)$$

Substituting $i = 1$ in equation (3.86), cubic spline in the 1st interval $[x_0, x_1] = [35, 45]$:

$$\begin{aligned} f_1(x) = & \frac{(45-x)^3}{60} \cdot 0 + \frac{(x-35)^3}{60} \cdot (-0.0878) + \frac{(45-x)}{10} \left(2.4 - \frac{100}{6} \cdot 0 \right) \\ & + \frac{(x-35)}{10} \left(12.4 - \frac{100}{6} \cdot (-0.0878) \right) \end{aligned} \quad (3.103)$$

Substituting $i = 2$ in equation (3.86), cubic spline in the 2nd interval $[x_1, x_2] = [45, 55]$:

$$\begin{aligned} f_2(x) = & \frac{(45-x)^3}{60} \cdot 0 + \frac{(x-35)^3}{60} \cdot (-0.0878) + \frac{(45-x)}{10} \left(2.4 - \frac{100}{6} \cdot 0 \right) \\ & + \frac{(x-35)}{10} \left(12.4 - \frac{100}{6} \cdot (-0.0878) \right) \end{aligned} \quad (3.104)$$

Substituting $i = 3$ in equation (3.86), cubic spline in the 3rd interval $[x_2, x_3] = [55, 65]$:

$$\begin{aligned} f_3(x) = & \frac{(65-x)^3}{60} \cdot 0.0093 + \frac{(x-55)^3}{60} \cdot 0.0745 + \frac{(65-x)}{10} \left(16.7 - \frac{100}{6} \cdot 0.0093 \right) \\ & + \frac{(x-55)}{10} \left(21.4 - \frac{100}{6} \cdot 0.0745 \right) \end{aligned} \quad (3.105)$$

Substituting $i = 4$ in equation (3.86), cubic spline in the 4th interval $[x_3, x_4] = [65, 75]$:

$$\begin{aligned} f_4(x) = & \frac{(75-x)^3}{60} \cdot 0.0745 + \frac{(x-65)^3}{60} \cdot (-0.2715) + \frac{(75-x)}{10} \left(21.4 - \frac{100}{6} \cdot 0.0745 \right) \\ & + \frac{(x-65)}{10} \left(26.7 - \frac{100}{6} \cdot (-0.2715) \right) \end{aligned} \quad (3.106)$$

Substituting $i = 5$ in equation (3.86), we get the cubic spline in the 5th interval $[x_4, x_5] = [75, 85]$:

$$f_5(x) = \frac{(85-x)^3}{60} \cdot (-0.2715) + \frac{(x-75)^3}{60} \cdot 0.0334 + \frac{(85-x)}{10} \left(26.7 - \frac{100}{6} \cdot (-0.2715) \right) + \frac{(x-75)}{10} \left(15.7 - \frac{100}{6} \cdot 0.0334 \right) \quad (3.107)$$

Substituting $i = 6$ in equation (3.86), we get the cubic spline in the 6th interval $[x_5, x_6] = [85, 95]$:

$$f_6(x) = \frac{(95-x)^3}{60} \cdot 0.0334 + \frac{(x-85)^3}{60} \cdot 0 + \frac{(95-x)}{10} \left(15.7 - \frac{100}{6} \cdot 0.0334 \right) + \frac{(x-85)}{10} \left(2.4 - \frac{100}{6} \cdot 0 \right) \quad (3.108)$$

Original Equations:

$$S0(x) = -0.0016 * (x - 35.0000)^3 + 0.0000 * (x - 35.0000)^2 + 1.1552 * (x - 35.0000) + 2.4000$$

$$S1(x) = 0.0021 * (x - 45.0000)^3 + -0.0465 * (x - 45.0000)^2 + 0.6897 * (x - 45.0000) + 12.4000$$

$$S2(x) = -0.0006 * (x - 55.0000)^3 + 0.0152 * (x - 55.0000)^2 + 0.3762 * (x - 55.0000) + 16.7000$$

$$S3(x) = 0.0005 * (x - 65.0000)^3 + -0.0022 * (x - 65.0000)^2 + 0.5057 * (x - 65.0000) + 21.4000$$

$$S4(x) = 0.0001 * (x - 75.0000)^3 + 0.0118 * (x - 75.0000)^2 + 0.6011 * (x - 75.0000) + 26.7000$$

$$S5(x) = 0.0000 * (x - 8.0000)^3 + -0.0040 * (x - 8.0000)^2 + 0.0805 * (x - 8.0000) + 15.7000$$

Final Equations:

$$S0(x) = -0.0016 * x^3 + 0.1629 * x^2 + -4.5472 * x + 28.4967 \quad (3.109)$$

$$S1(x) = 0.0021 * x^3 + -0.3244 * x^2 + 17.3834 * x + -300.4627 \quad (3.110)$$

$$S2(x) = -0.0006 * x^3 + 0.1112 * x^2 + -6.5738 * x + 138.7538 \quad (3.111)$$

$$S3(x) = 0.0005 * x^3 + -0.0935 * x^2 + 6.7311 * x + -149.5187 \quad (3.112)$$

$$S4(x) = 0.0001 * x^3 + -0.0059 * x^2 + 0.1600 * x + 14.7583 \quad (3.113)$$

$$S5(x) = 0.0000 * x^3 + -0.0044 * x^2 + 0.1478 * x + 14.7907 \quad (3.114)$$

??.

3.3 Insert Table

Table 3.1: Data

Age	Death percentage%
31-40	2.4
41-50	12.4
51-60	16.7
61-70	21.4
71-80	26.7
81-90	15.7
91-100	2.4

Table 3.2: Mean value table

Mean Age	Death percentage
35	2.4
45	12.4
55	16.7
65	21.4
75	26.7
8	15.7
95	2.4

Below are the pie charts and bar charts that were used to obtain information related to our research.

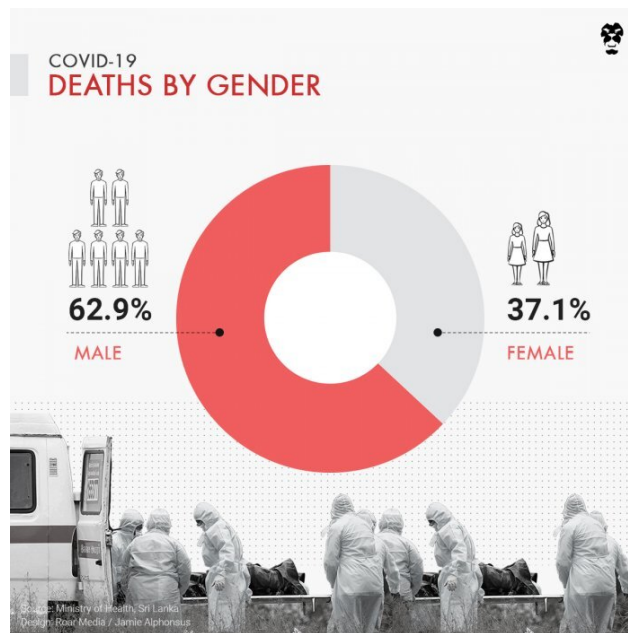


Figure 3.1: - Covid-19 Deaths by Gender

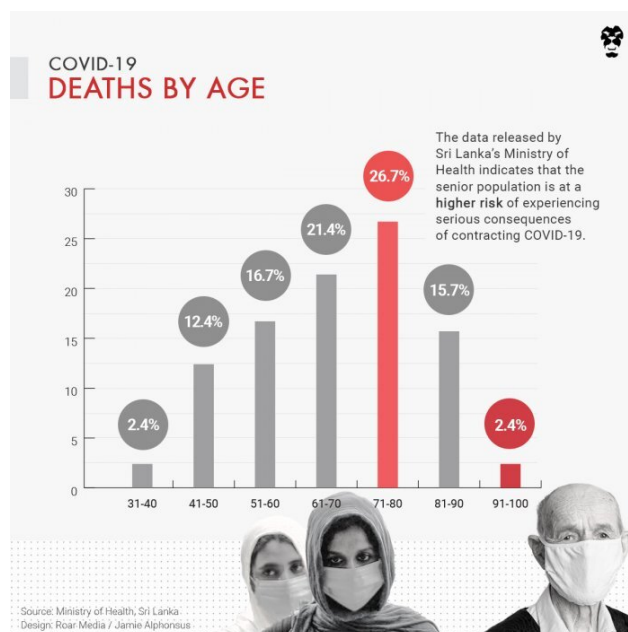


Figure 3.2: - Covid-19 Deaths by Age

Chapter 4

Results

Original equations:

```
S0(x) = -0.0015 * (x - 35.0000)^3 + 0.0000 * (x - 35.0000)^2 + 1.1464 * (x - 35.0000) + 2.4000
S1(x) = 0.0016 * (x - 45.0000)^3 + -0.0439 * (x - 45.0000)^2 + 0.7072 * (x - 45.0000) + 12.4000
S2(x) = 0.0011 * (x - 55.0000)^3 + 0.0047 * (x - 55.0000)^2 + 0.3147 * (x - 55.0000) + 16.7000
S3(x) = -0.0058 * (x - 65.0000)^3 + 0.0373 * (x - 65.0000)^2 + 0.7340 * (x - 65.0000) + 21.4000
S4(x) = 0.0051 * (x - 75.0000)^3 + -0.1357 * (x - 75.0000)^2 + -0.2507 * (x - 75.0000) + 26.7000
S5(x) = -0.0006 * (x - 85.0000)^3 + 0.0167 * (x - 85.0000)^2 + -1.4412 * (x - 85.0000) + 15.7000
```

Figure 4.1: - Original equations

Final output:

```
S0(x) = -0.0015 * x^3 + 0.1537 * x^2 + -4.2333 * x + 25.0389
S1(x) = 0.0016 * x^3 + -0.2625 * x^2 + 14.4964 * x + -255.9064
S2(x) = 0.0011 * x^3 + -0.1747 * x^2 + 9.6658 * x + -167.3438
S3(x) = -0.0058 * x^3 + 1.1618 * x^2 + -77.2068 * x + 1714.8938
S4(x) = 0.0051 * x^3 + -1.2789 * x^2 + 105.8481 * x + -2861.4764
S5(x) = -0.0006 * x^3 + 0.1585 * x^2 + -16.3323 * x + 600.2989
```

Figure 4.2: - Final output

Cubic spline graph

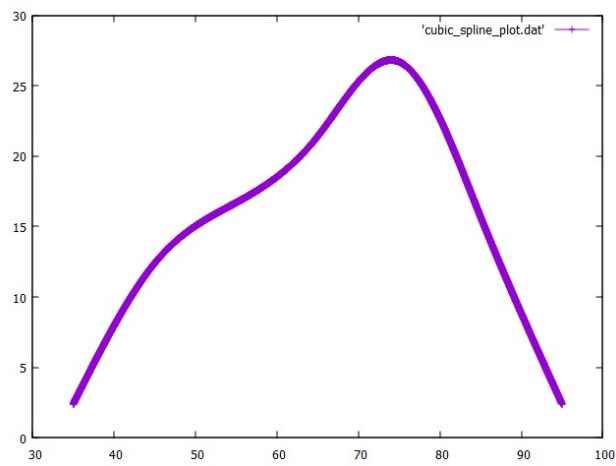


Figure 4.3: - Cubic spline graph

Chapter 5

Discussion

- Although the age factor has prevailed in the death rates all over the world, it has brought different results in different world areas because of the dissimilarities of the health care systems, public health insurance policies, and socio-economic factors. For instance: With more developed healthcare systems, it is apparent that countries such as the United States or countries in Europe have been experiencing notable mortality, particularly in the elderly population, yet vaccination and access to critical care have been able to decrease some of the effects. especially those countries and areas with scarce health care facilities; for instance, India and some parts of Africa have been overwhelmed with high case loads. At times, younger generation populations have also suffered a higher mortality rate because of issues such as inadequate or delayed access to health care or initial health risks such as malnutrition or infectious diseases.
- The emergence of COVID-19 variants, such as Delta and Omicron, has had varying effects on mortality across age groups: The new strains of COVID, such as delta and omicron, have affected mortality rates in different ways by age brackets. It was associated with a higher prevalence, severe illness, and death rate in persons with poor health and the elderly. In like manner, other populations, particularly the young, also experienced a shift up in severity of disease as opposed to previous variants. The while more transmissible, generally infected people with less severe illness leading to a lesser mortality rate but not safe for unvaccinated elderly people and individuals having compromised immune systems.
- The emergence of different types of COVID-19, such as the delta and omicron types, have different relative death rates among age groups, including: : Increased severity

of infections and disease is worse for older adults and people with various medical conditions. In addition, the disease is more dangerous for younger populations than for older species. is more contagious but generally less contagious and less deadly. But it's still a big problem for unvaccinated seniors and patients with chronic illnesses.

- The understanding of age-based mortality risk has shaped a wide range of public health policies. These insights of age-based per capita mortality risk have influenced numerous types of public health policies.
- Initially, there was drastic implementation of the quarantine to minimize contact and interaction so as to reduce endangering contacts, especially to elderly individuals. But these measures also cost in the social, psychological, and even financial sense to the young people, as the discussion shifted about how one can shield the vulnerable groups without necessarily putting pressure on the larger society.
- While children did not experience high mortality rates, the schools were shut down for a long time as a preventive measure, intending to avoid the formation of crowds of corona positive people and to protect the old and the sick individuals. There are, however, some concerns in terms of the development, education, and even social effects of these devices on children and adolescents. Vaccinations were at first focused on the elderly, people working in the health care sector, and the remaining phases of the community. While this strategy contributed positively to the reduction of deaths, it also brought to light other facets with regard to the equitable distribution of the vaccine the world over.
- Pandemic has raised ethical questions about how the society can safeguard the vulnerable while being fair and equal. Key ethical discussions include.
- Some communities found themselves in having to physically restrain the distribution of ventilators and ICU beds at the peak of the outbreak, leading to important questions as to whether patient age should be used as a criterion for rationing access to treatments.
- Younger generations depicted a grim picture of life and work disruptions caused by child virus mitigation measures leading to discussions on such issues as protection of elders as compared to when young individuals, especially students and employees must sit at home to learn or work respectively.

Chapter 6

Conclusion

- • There are always higher COVID-19 mortalities for those over sixty-five years, meaning that they bear a higher risk as compared to the youthful population. This is because of predisposing factors of overall health in clients and a reduced immunity to diseases in old age. Children and young adults produce comparatively negligible signs and deaths, however, if confirmed by diseases or weak immune systems then it will need different consideration. Promotions such as the vaccination campaign and priority given to the isolation of elderly adults are aimed at risky persons. This particular division of mortality based on age is going to be helpful in dealing with future pandemics further.
- A plethora of studies conducted by researchers from all across the world show that age is a major risk factor causing the worse prognosis of COVID-19 infection, as the mortality rate among elderly people is exceptionally higher as compared to that among other population groups, particularly those persons who are 70 or 80 years old and above. Immunization is the property of older age, and comorbidity factors, for example, hypertension, diabetes, cardiovascular disease, and chronic respiratory disease, increase the risk of severe COVID-19 and death.
- While death rates among the middle aged have declined, there is still the chance of a fatal encounter, and often this is not just one fatal experience but multiple fatal experiences for those with multiple diseases. On the other hand, an overall healthy young population of age between 20 and 50 years, thus having lower morbid and mortal rates, sometimes even experiences severe disease.
- As such, children and adolescents' death rate has been exceptionally low compared

to the death rate of adults. In total, young people are getting infected with COVID-19, and in the vast majority, they either show mild signs or none at all. But there is something that has been observed very scarcely; it is referred to as Multisystem Inflammatory Syndrome in Children (MIS-C).

- COVID-19 mortality depends on age and the preexisting health conditions whereby people aged 65 years or older have reduced immune capacity to fight the virus. Children and young adults have mild diseases and low mortality rates as compared with older adults. Measures such as vaccination campaigns and the use of face masks are aimed at endangered groups of people – the elderly. The severity and the threat of COVID-19 will depend on these age distributions and thus, future management of subsequent pandemics has to take them into account.

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Appendices

(C code)

Appendix A

Main C code

```
# Remove first two columns from seqdata
countdata <- data[,-1]
row.names(countdata)<-data[,1]
```

```
1
2
3 #include <stdio.h>
4 #include <stdlib.h>
5
6 void TridiagonalEquationsSolver(int n, double *a, double *b, double *c, double
   *d, double *x) {
7     double alpha[n], beta[n];
8
9     // Forward substitution
10    alpha[0] = b[0];
11    beta[0] = d[0] / b[0];
12
13    for (int i = 1; i < n; i++) {
14        alpha[i] = b[i] - a[i] * c[i - 1] / alpha[i - 1];
15        beta[i] = (d[i] - a[i] * beta[i - 1]) / alpha[i];
16    }
17
18    // Backward substitution
19    x[n - 1] = beta[n - 1];
20    for (int i = n - 2; i >= 0; i--) {
```

```

21     x[i] = beta[i] - c[i] * x[i + 1] / alpha[i];
22 }
23 }
24
25 int main() {
26     int n;
27
28     printf("Enter number of data points: ");
29     scanf("%d", &n);
30
31     double x[n], y[n], h[n - 1], a[n - 1], b[n], c[n - 1], d[n - 1], M[n];
32
33     // Read data points
34     for (int i = 0; i < n; i++) {
35         printf("Enter x%d and y%d: ", i + 1, i + 1);
36         scanf("%lf %lf", &x[i], &y[i]);
37     }
38
39     // Calculate h[i]
40     for (int i = 0; i < n - 1; i++) {
41         h[i] = x[i + 1] - x[i];
42     }
43
44     // Set up the tridiagonal system
45     for (int i = 1; i < n - 1; i++) {
46         a[i] = h[i - 1];
47         b[i] = 2 * (h[i - 1] + h[i]);
48         c[i] = h[i];
49         d[i] = 6 * ((y[i + 1] - y[i]) / h[i] - (y[i] - y[i - 1]) / h[i - 1]);
50     }
51
52     // Solve the system
53     TridiagonalEquationsSolver(n - 2, a + 1, b + 1, c + 1, d + 1, M + 1);
54
55     // Set boundary conditions
56     M[0] = 0;
57     M[n - 1] = 0;
58
59     // Open file to save the results
60     FILE *fp = fopen("cubic_spline_plot.dat", "w");
61
62     // Output the original spline equation and write data points to the file
63     double A[n - 1], B[n - 1], C[n - 1], D[n - 1];

```

```

64  for (int i = 0; i < n - 1; i++) {
65      A[i] = (M[i + 1] - M[i]) / (6 * h[i]);
66      B[i] = M[i] / 2;
67      C[i] = (y[i + 1] - y[i]) / h[i] - (M[i + 1] + 2 * M[i]) * h[i] / 6;
68      D[i] = y[i];
69
70      // Output the cubic spline equations
71      printf("S%d(x) = %.4f * (x - %.4f)^3 + %.4f * (x - %.4f)^2 + %.4f * (x
- %.4f) + %.4f\n",
72          i, A[i], x[i], B[i], x[i], C[i], x[i], D[i]);
73
74      // Write the data points for this segment to the file
75      double xi = x[i];
76      while (xi <= x[i + 1]) {
77          double yi = A[i] * (xi - x[i]) * (xi - x[i]) * (xi - x[i])
78              + B[i] * (xi - x[i]) * (xi - x[i])
79              + C[i] * (xi - x[i])
80              + D[i];
81          fprintf(fp, "%.4f %.4f\n", xi, yi);
82          xi += 0.01;
83      }
84  }
85
86  fclose(fp);
87
88  // Final output - transform to a standard polynomial form
89  printf("\nFinal output in polynomial form:\n\n");
90  for (int i = 0; i < n - 1; i++) {
91      double a_poly = A[i];
92      double b_poly = B[i] - 3 * A[i] * x[i];
93      double c_poly = C[i] + 3 * A[i] * x[i] * x[i] - 2 * B[i] * x[i];
94      double d_poly = D[i] - A[i] * x[i] * x[i] * x[i] + B[i] * x[i] * x[i]
- C[i] * x[i];
95
96      // Output the transformed cubic polynomial
97      printf("S%d(x) = %.4f * x^3 + %.4f * x^2 + %.4f * x + %.4f\n",
98          i, a_poly, b_poly, c_poly, d_poly);
99  }
100
101  printf("\nData points have been written to 'cubic_spline_plot.dat'.\n");
102
103  // User interaction for substituting x_value
104  double x_value, result;

```



```
105
106 while (1) {
107     printf("\nEnter the x value to substitute (or enter -1 to exit): ");
108     scanf("%lf", &x_value);
109
110     if (x_value == -1) {
111         printf("Exiting...\n");
112         break;
113     }
114
115     // Automatically determine which spline equation to use
116     int eq_choice = -1;
117     for (int i = 0; i < n - 1; i++) {
118         if (x_value >= x[i] && x_value <= x[i + 1]) {
119             eq_choice = i;
120             break;
121         }
122     }
123
124     if (eq_choice == -1) {
125         printf("x value is out of range. Please try again.\n");
126         continue;
127     }
128
129     // Evaluate the chosen spline equation
130     result = A[eq_choice] * (x_value - x[eq_choice]) * (x_value - x[
eq_choice]) * (x_value - x[eq_choice])
131         + B[eq_choice] * (x_value - x[eq_choice]) * (x_value - x[
eq_choice])
132         + C[eq_choice] * (x_value - x[eq_choice])
133         + D[eq_choice];
134
135     printf("S%d(%.4f) = %.4f\n", eq_choice, x_value, result);
136 }
137
138 return 0;
139 }
```